

840065

Course Name: Digital Image Processing**Course Outcome**

CO1- Understand mathematical formulation of an image, its processing steps and relationship between image pixels.

CO2- Apply Image enhancement using intensity transformations and spatial filtering.

CO3- Analyze image enhancement for frequency domain using Fourier transform.

CO4- Formulate region of interest through morphological operations.

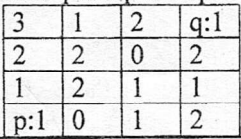
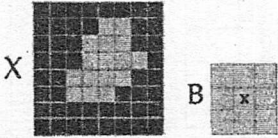
CO5- Evaluate strongly co-related regions obtained through Segmentation using discontinuity and homogeneity-based segmentation techniques

CO6: Describe an object of an image using Shape Number and Boundary descriptors.

Printed Pages:**University Roll No.****End Term Examination, Odd Semester 2025-26****B.Tech CS, III Year, V Semester****Subject Code & Subject Name- BCSE 0101 & Digital Image Processing****Time: 3 Hours****Maximum Marks: 50****Instruction for students:**

1. All parts of a question should be answer at one place.
2. First complete the one set of section then start a new section.

Section – A**Attempt All Questions****4 X 5 = 20 Marks**

No.	Detail of Question	Marks	CO	BL	KL
1	Explain the fundamental steps in digital image processing with the help of block diagram.	4	1	U	P
2	Consider an image segment as shown in Fig. Let $V = \{1, 2\}$. Compute the length of shortest 4 path and 8 path between pixel p and q. 	4	2	C	M
3	Compute the 1D-DFT of the 4x4 grayscale image using twiddle factor. $f(x) = [1, 1, 1, 1]$	4	3	E	P
4	Perform dilation and erosion operation on the following image. 	4	4	U	P
5	Apply the split and merge technique for the given image by considering the threshold value 4.	4	5	U	C

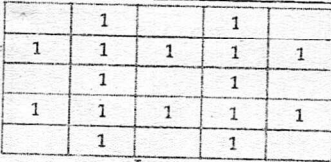
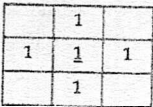
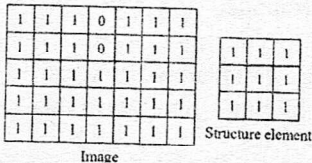
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		1	0	1	0	2	3	5	4					
		Or												
		Imagine you are developing an intelligent traffic surveillance system that needs to detect vehicles on roads in real-time. The system cameras operate even at night, in low-light conditions, and sometimes in fog. To achieve accurate vehicle detection, you must extract clear boundaries from the captured images before passing them to the object detection model. In this scenario, how effective would the Canny Edge Detection Algorithm be? Explain its working steps and key advantages.												

Section – B

Attempt All Questions

3 X 5 = 15 Marks

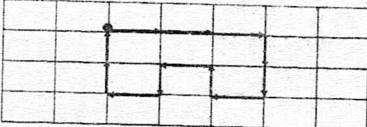
No.	Detail of Question	Marks	CO	BL	KL																									
6	What are the different thresholding methods for segmentation. Write down the basic global thresholding algorithm.	3	5	U	C																									
7	Consider the image shown in Fig. Show the result of the region growing algorithm. Let the threshold ≤ 4 and the seed value 9. <table><tr><td>1</td><td>0</td><td>7</td><td>8</td><td>7</td></tr><tr><td>0</td><td>1</td><td>8</td><td>9</td><td>8</td></tr><tr><td>0</td><td>0</td><td>7</td><td>9</td><td>8</td></tr><tr><td>0</td><td>1</td><td>8</td><td>8</td><td>9</td></tr><tr><td>1</td><td>2</td><td>8</td><td>8</td><td>9</td></tr></table>	1	0	7	8	7	0	1	8	9	8	0	0	7	9	8	0	1	8	8	9	1	2	8	8	9	3	5	An	C
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8	You are working with a 5×5 binary image which contains a pattern and a structuring element. After applying hit or miss transform on the image, what will be the output of the image?	3	4	R	C																									

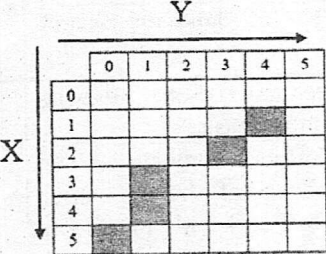
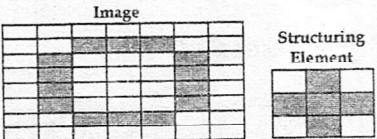
	 Image	 Structuring Element				
9	Define Edge detection technique (Sobel, Prewitt, Roberts) along with the suitable mask.	3	5	An	P	
10	You are working on a digital pathology system that analyses microscopic images of human tissue samples. The goal is to detect and study the shapes of individual cells to identify abnormal cell growth patterns. While processing the tissue image, you notice that many cells overlap or cluster together. Identify the internal boundaries of each cell to separate individual cell structures for the following image.  Image	3	4	E	P	

Section – C

Attempt All Questions

5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Find the following shape description of the given shape (black circle is the origin and move in the clockwise direction) using 4 directional chain code a) Chain Code b) First difference c) Circular First Difference d) Shape Number e) Order 	5	6	An	C

12	Join the given set of points in the following image using Hough Transform with the given conditions. $-2 \leq m \leq 2$ and step size is 1.0. 	5	5	An	C
13	In a medical image application, you are given a binary image of a cross-section scan where tumor regions appear as enclosed black areas surrounded by white tissue structures. Some of these black regions are completely surrounded and some are connected to background. You are asked to automatically fill only the completely enclosed black region. Apply the algorithm and show the step-wise output. 	5	4	U	C